UNIT 3 · TOPIC 6(A)

Field Capacity and Field Efficiency

Syllabus topic: Calculations of field capacity, efficiency and rates of seed, fertilizer and chemical applicators

Definitions and formulas

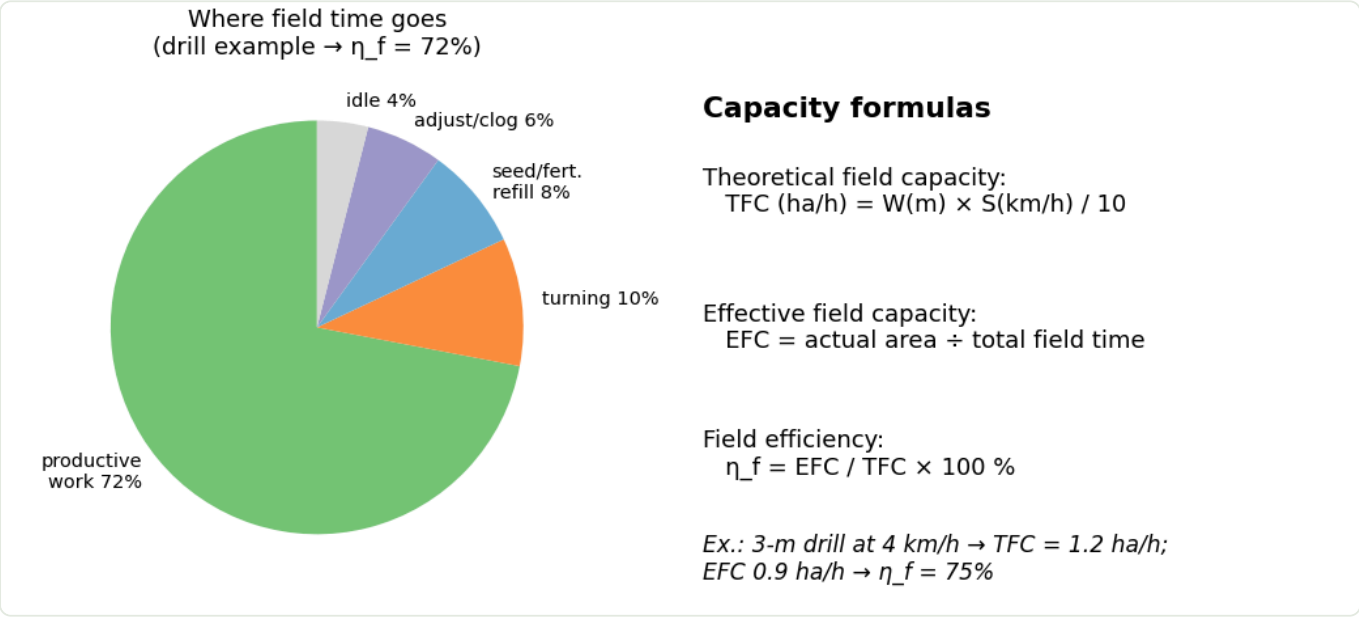
$$\text{Theoretical field capacity TFC (ha/h)} = W \times S / 10$$

W = rated working width (m); S = forward speed (km/h). It assumes 100% of the width and 100% of the time are used.

- **Effective field capacity (EFC)** = actual area covered ÷ total field time (including turning, refilling, unloading, adjustment, minor clogging).
- **Material capacity** (kg/h or t/h) is used for combines/threshers/balers, where throughput — not area — limits the work.

$$\text{Field efficiency } \eta_f = EFC/TFC \times 100$$

Field efficiency is NOT constant for a machine: it falls with small/irregular fields (more turning), full-width overlap and frequent hopper/tank service.



Where field time goes (the pie illustrates a 72% case; the worked example below gives 75%)

Operation	Typical field efficiency
Ploughing	75–85%
Disc harrowing	80–90%
Seed drilling	65–85%
Combine harvesting	65–80%
Boom spraying	50–65% (refill-dominated)
Rotavator	80–90%

Worked examples

Ex.1 3-m seed drill at 4 km/h → TFC = 3 × 4/10 = 1.2 ha/h; measured EFC = 0.9 ha/h → **η_f = 75%**.

Ex.2 4.2-m combine at 5 km/h, η_f = 70% → EFC = (4.2 × 5/10) × 0.70 = **1.47 ha/h**; a 25-ha block needs 25/1.47 ≈ **17 h**.

Ex.4 (days to finish) 60 ha to drill, EFC 1.0 ha/h, 8 h/day → 7.5 → **8 days**. Timeliness planning (Unit 2, Topic 3) decides whether a second drill is needed.

Practice MCQs

Q71. Theoretical field capacity (ha/h) of a machine of width W (m) at speed S (km/h) is:

(a) WS (b) WS/10
(c) WS/100 (d) 10WS

Q72. A 3-m drill at 4 km/h has TFC of:

(a) 0.75 ha/h (b) 1.2 ha/h
(c) 12 ha/h (d) 2.4 ha/h

Q73. If its effective capacity is 0.9 ha/h, field efficiency is:

(a) 60% (b) 70%
(c) 75% (d) 90%

Q74. Field efficiency accounts for:

(a) Only speed loss (b) Turning, refilling, overlap and idle time
(c) Fuel quality (d) Soil texture only

Q75. A 4.2-m combine at 5 km/h with 70% field efficiency covers per hour about:

(a) 1.05 ha (b) 1.47 ha
(c) 2.1 ha (d) 2.94 ha

Q76. Time to combine 25 ha at EFC 1.47 ha/h ≈

(a) 12 h (b) 17 h
(c) 25 h (d) 34 h

Q77. The field operation typically having the LOWEST field efficiency is:

(a) Ploughing (b) Harrowing
(c) Boom spraying (frequent refills) (d) Rotavation

Answers

Q71 — (b) WS/10
Q72 — (b) 1.2 ha/h — *TFC = 3×4/10 = 1.2 ha/h.*
Q73 — (c) 75% — *η_f = 0.9/1.2 = 75%.*
Q74 — (b) Turning, refilling, overlap and idle time
Q75 — (b) 1.47 ha — *EFC = (4.2×5/10)×0.7 = 1.47 ha/h.*
Q76 — (b) 17 h — *25/1.47 = 17 h.*
Q77 — (c) Boom spraying (frequent refills)

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities) & Ch. 17; RNAM test codes (combine, thresher, transplanter); BIS standards for power threshers and chaff cutters; Jagdishwar Sahay; NPTEL lectures.